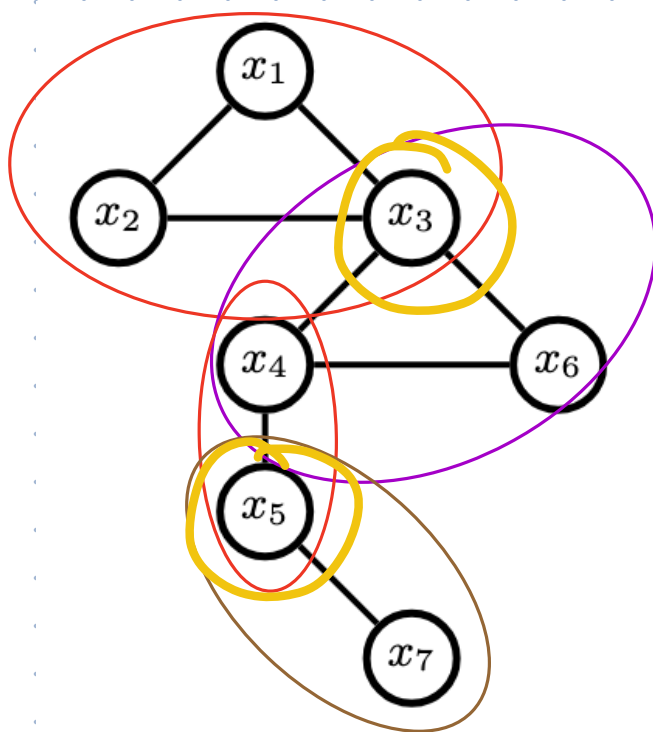


Q1.1

(a)



MRF:

$$p(x_1, \dots, x_7) = \frac{1}{Z} \phi_1(x_1, x_2, x_3) \phi_2(x_3, x_4, x_6) \phi_3(x_4, x_5) \phi_4(x_5, x_7)$$

(b)

Given $S = \{x_3, x_5\}$

$$A = \{x_1, x_2\}$$

$$B = \{x_4, x_6, x_7\}$$

$$A = \{x_4, x_6\}$$

$$B = \{x_1, x_2, x_7\}$$

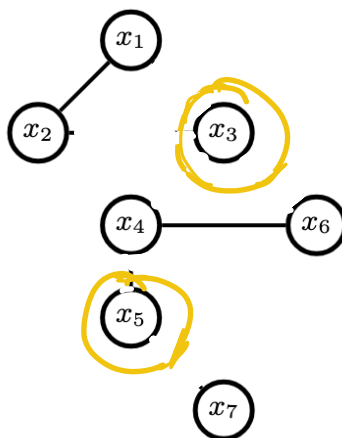
$$A = \{x_7\}$$

$$B = \{x_1, x_2, x_4, x_6\}$$

Blanket of x_4 :

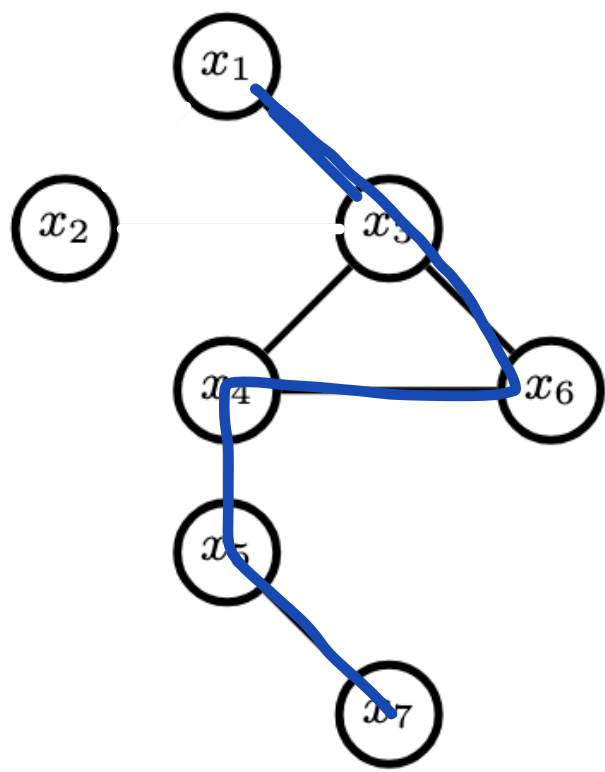
↳ Neighbours of x_4 :

x_5, x_6, x_3

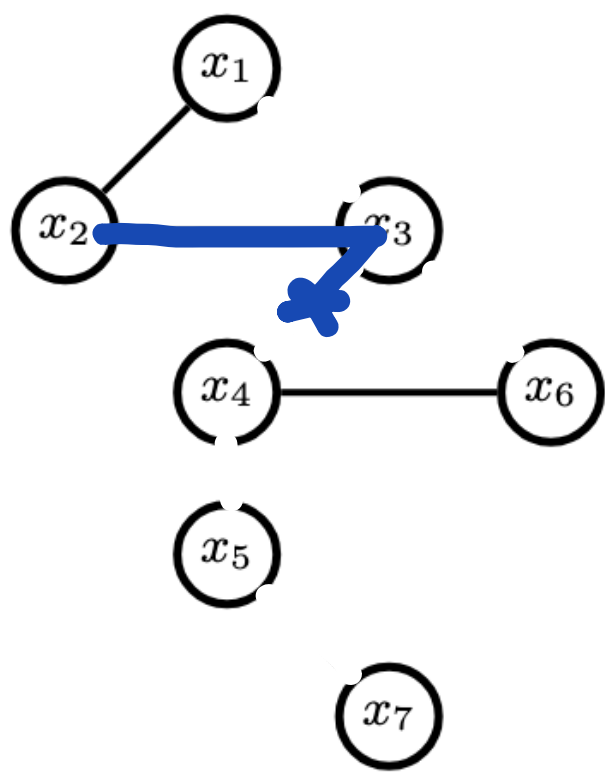


(9)

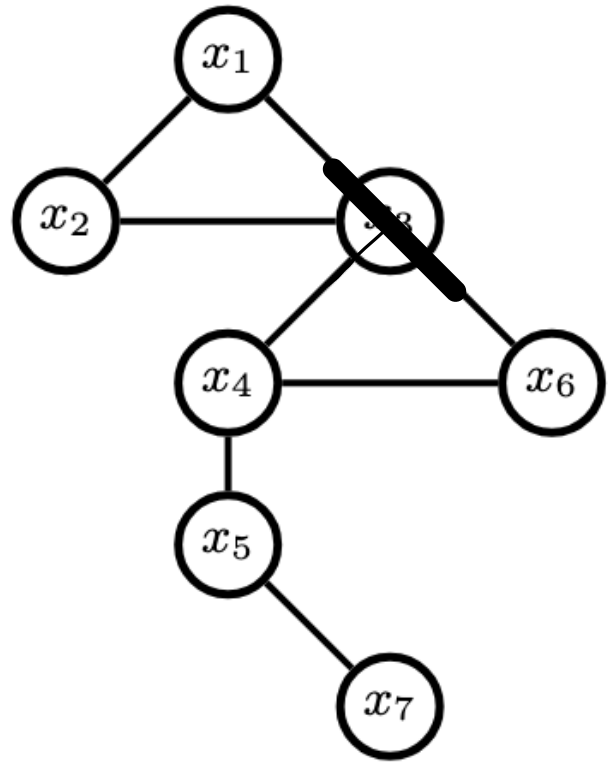
• $x_1 || x_5 | x_2$
false



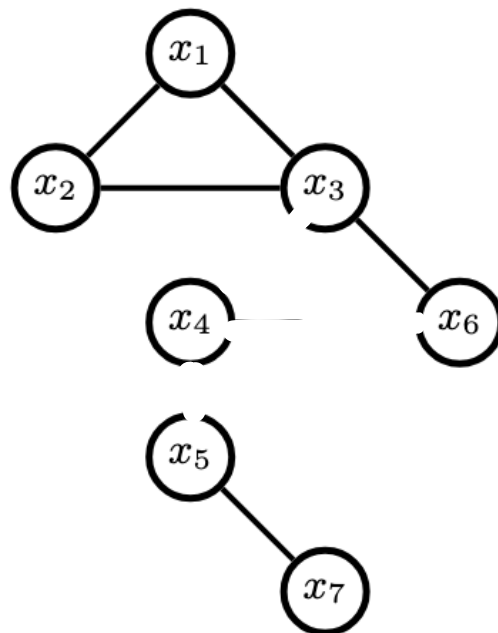
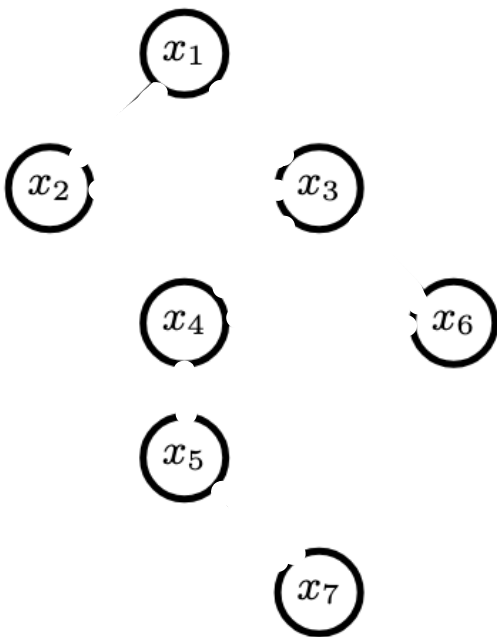
• $x_2 \perp\!\!\!\perp x_7 | \{x_3, x_5\}$
True



- Marginalizing over x_3 makes x_1 & x_4 dependent
True



- $P(x_4 \mid x_1, x_2, x_3, x_4, x_5, x_6) = P(x_4 \mid x_3, x_5, x_6)$



True

(c) Prove

$$p(a, b | c) = p(a | c) p(b | c)$$

LHS:

$$\frac{p(a, b, c)}{p(c)}$$

$$\frac{p(a, b, c)}{\sum_{a, b} p(a, b, c)}$$

$$\frac{\cancel{\frac{1}{Z}} \phi_1(a | c) \phi_2(b | c)}{\sum_{a, b} \cancel{\frac{1}{Z}} \phi_1(a | c) \phi_2(b | c)}$$

RHS:

$$\frac{p(a | c)}{p(c)} \times \frac{p(b | c)}{p(c)}$$

$$\frac{\sum_b p(a, b | c)}{\sum_{a, b} p(a, b | c)} \frac{\sum_a p(a, b | c)}{\sum_{a, b} p(a, b | c)}$$

$$\frac{\sum_b \cancel{\frac{1}{Z}} \phi_1(a | c) \phi_2(b | c)}{\sum_{a, b} \cancel{\frac{1}{Z}} \phi_1(a | c) \phi_2(b | c)} \frac{\sum_a \cancel{\frac{1}{Z}} \phi_1(a | c) \phi_2(b | c)}{\sum_{a, b} \cancel{\frac{1}{Z}} \phi_1(a | c) \phi_2(b | c)}$$

Now,

$$\frac{\phi_1(a, c) \phi_2(b, c) = \sum_b \phi_1(a, c) \phi_2(b, c) \sum_a \phi_1(a, c) \phi_2(b, c)}{\sum_{a, b} \phi_1(a, c) \phi_2(b, c)}$$

Let,

$$\phi_1(a, c) = [a == c] \quad \phi_2(b, c) = [b == c]$$

$$\begin{aligned} \underbrace{\phi_1(a, c) \phi_2(b, c)}_{[a == c][b == c]} &= \underbrace{\sum_{a, b} \phi_1(a, c) \phi_2(b, c)}_1 \\ &= \underbrace{\sum_b \phi_1(a, c) \phi_2(b, c)}_{[a == c]} \underbrace{\sum_a \phi_1(a, c) \phi_2(b, c)}_{[b == c]} \end{aligned}$$

Note:

I believe this is correct but solution manual is different. So, I can't be sure of its correctness as I have no TA :)

Okay, I realized that this proof is not general. This would have worked if I wanted to show uniqueness by contradiction.

lets try again,

RHS: $\frac{p(a,c)}{p(c)} \times \frac{p(b,c)}{p(c)}$

$$\frac{\sum_b p(a,b,c)}{\sum_{a,b} p(a,b,c)} \frac{\sum_a p(a,b,c)}{\sum_{a,b} p(a,b,c)}$$

$$\frac{\sum_b \cancel{\frac{1}{Z}} \phi_1(a,c) \phi_2(b,c)}{\sum_{a,b} \cancel{\frac{1}{Z}} \phi_1(a,c) \phi_2(b,c)} \frac{\sum_a \cancel{\frac{1}{Z}} \phi_1(a,c) \phi_2(b,c)}{\sum_{a,b} \cancel{\frac{1}{Z}} \phi_1(a,c) \phi_2(b,c)}$$

$$\frac{\sum_b \phi_1(a,c) \phi_2(b,c)}{\left[\sum_{a,b} \phi_1(a,c) \phi_2(b,c) \right]^2} \frac{\sum_a \phi_1(a,c) \phi_2(b,c)}{\left[\sum_{a,b} \phi_1(a,c) \phi_2(b,c) \right]^2}$$

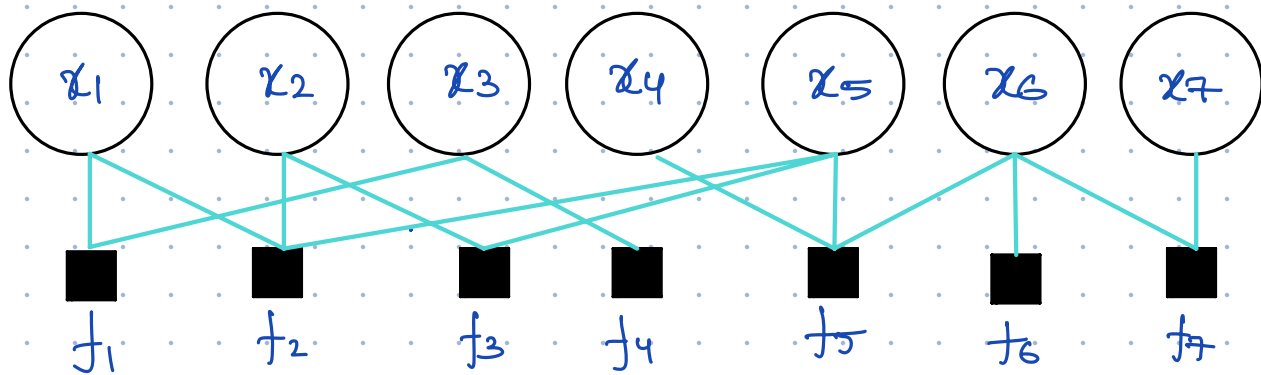
$$\frac{\phi_1(a,c) \sum_b \phi_2(b,c) \phi_2(b,c) \sum_a \phi_1(a,c)}{\left[\sum_{a,b} \phi_1(a,c) \phi_2(b,c) \right]^2}$$

$$\frac{\phi_1(a,c) \phi_2(b,c) \left[\cancel{\sum_b \phi_2(b,c)} \sum_a \phi_1(a,c) \right]}{\left[\sum_{a,b} \phi_1(a,c) \phi_2(b,c) \right]^2}$$

$$\frac{\phi_1(a,c) \phi_2(b,c)}{\sum_{a,b} \phi_1(a,c) \phi_2(b,c)} = \text{RHS}$$

Q1.2

(a)



(b) Given $\mathcal{X} = \{x_1, \dots, x_6\}$ with $x_i \in \{0, 1\} \forall i$, a Markov Random Field is given by the distribution

$$p(x_1, \dots, x_6) = \frac{1}{Z} \phi_1(x_1, x_3) \phi_2(x_1, x_2, x_5) \phi_3(x_4, x_5, x_6).$$

Define the potentials as

$$\begin{aligned} \phi_1(x_1, x_3) &= [x_1 = x_3] \\ \phi_2(x_1, x_2, x_5) &= [x_2 = x_5] \cdot [x_1 = 1] \\ \phi_3(x_4, x_5, x_6) &= 2x_4x_5 + 3x_6. \end{aligned}$$

$$Z = \sum_{x_1, x_2, x_3, x_4, x_5, x_6} \phi_1(x_1, x_3) \phi_2(x_1, x_2, x_5) \phi_3(x_4, x_5, x_6)$$

simply, use the combinations where $\phi_1(x_1, x_3) \phi_2(x_1, x_2, x_5)$ is 1 & plug in $\phi_3(x_4, x_5, x_6)$ & sum them up

Q 1.4

(a)

Advantages

↳ Integrate pixels

↳ Allows for certainty estimation

Drawbacks

↳ Computationally expensive

↳ Solutions are approximate

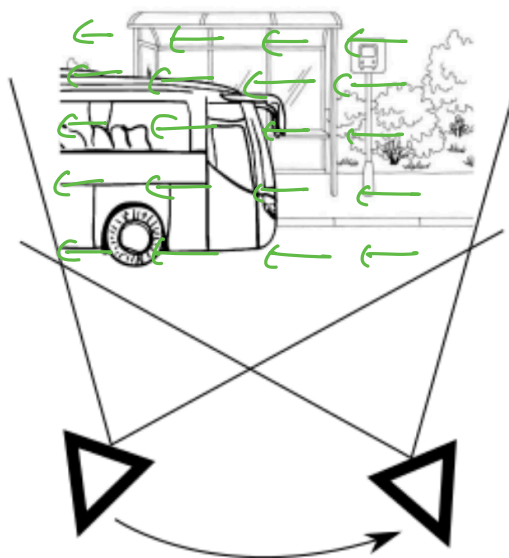
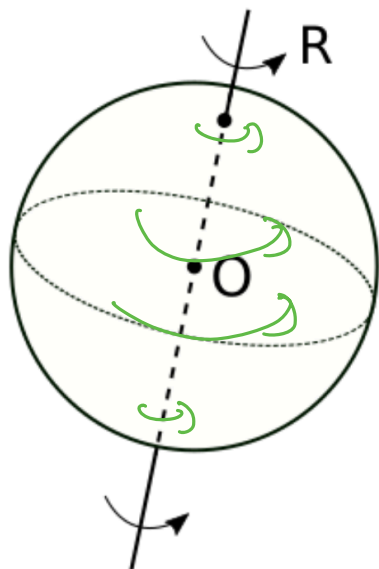
(b) Following assumption are made:

- Color constancy between pixels across images.

This is violated for:

- Non-lambertian objects
- Daylight changes

Q 1.5



motion field optical field

(b)



(c)

$$u_{x,y} \cdot I_x(x,y) + v_{x,y} \cdot I_y(x,y) + I_t(x,y) = 0$$

↳ Intensity remains through t
 We have two unknowns but only one equation (constraint)

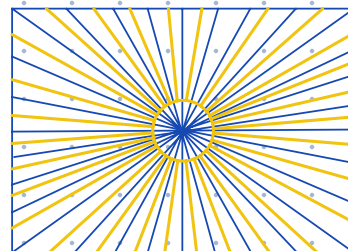
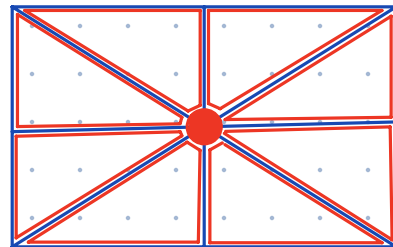
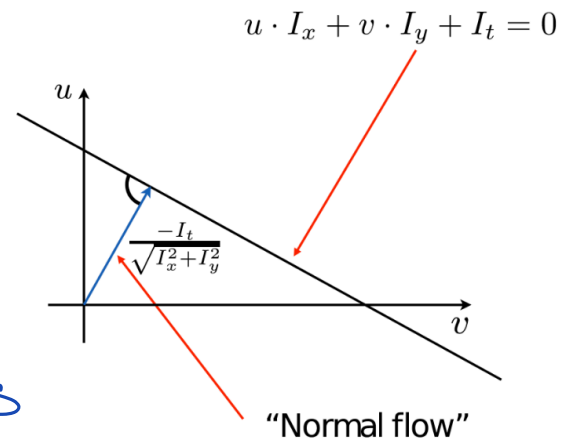
↳ The constraint says that brightness remains same i.e. $I_t = 0$
 So, we have

$$u I_x + v I_y = 0$$

$$(u, v) \cdot (I_x, I_y) = 0$$

Its direction is:

$$\hat{n} = \frac{(I_x, I_y)}{\sqrt{I_x^2 + I_y^2}}$$



we conclude that our motion direction is $\perp$ to $\hat{n}$.

To find, how far that motion is, we can assume d and say

$$d \cdot \hat{n} = 0$$

let $u = (\lambda dx) \quad v = (\lambda dy)$

$$I_x(\lambda dx) + I_y(\lambda dy) = -I_t$$

$$\lambda (I_x dx + I_y dy) = -I_t$$

$$\lambda = \frac{-I_t}{I_x dx + I_y dy}$$

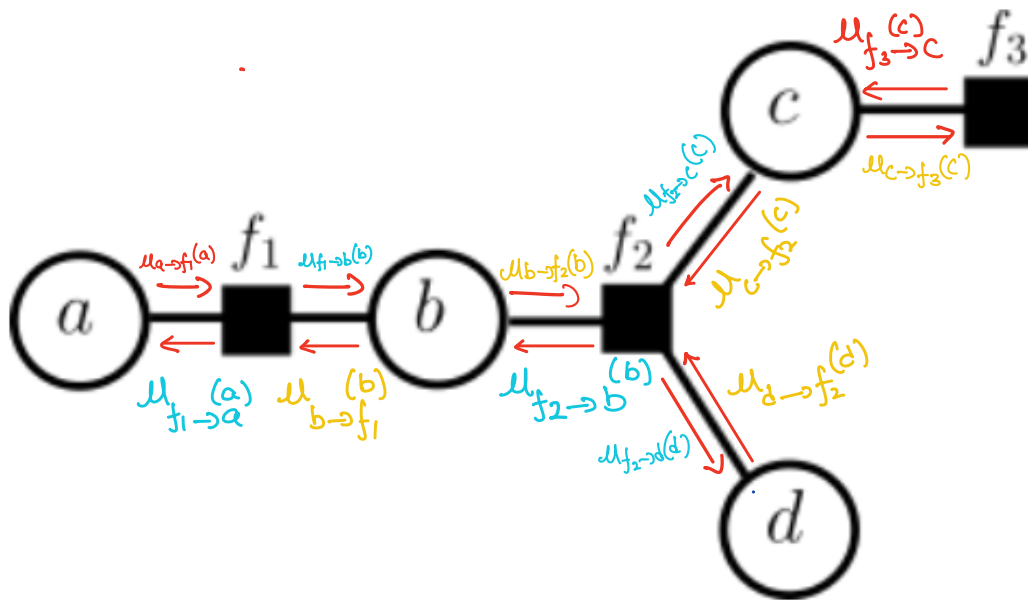
Choose any d such as

$$d = \frac{(-I_y, I_x)}{\sqrt{I_x^2 + I_y^2}}$$

Q1.3

"I was talking on the phone :)"

(b)



$$f_1(a, b) = [a = b], \quad f_2(b, c, d) = 0.6b + 0.2c + 0.3d$$

$$f_3(c) = [c = 1]$$

$$p(a, b, c, d) = \frac{1}{Z} f_1(a, b) f_2(b, c, d) f_3(c)$$

Now, Sum Product Algorithm:

1. Initialization:

$$\mu_{f_3 \rightarrow c}(c) = f_3(c) \quad \mu_{a \rightarrow f_1}(a) = 1$$

$$\mu_{d \rightarrow f_2}(d) = 1$$

2. Variable-to-factor Messages:

$$\mu_{c \rightarrow f_2}(c) = \prod_{g \in \{nc(c) \setminus f_2\}} \mu_{g \rightarrow c}(x)$$

$$= \prod_{g \in \{f_3\}} \mu_{g \rightarrow c}(x)$$

$$= \mu_{f_3 \rightarrow c}(c)$$

$$= f_3(c)$$

$$\mu_{c \rightarrow f_3}(c) = \prod_{g \in \{ne(c) \setminus f_3\}} \mu_{g \rightarrow c}(c)$$

$$= \prod_{g \in \{f_2\}} \mu_{g \rightarrow c}(c)$$

$$= \mu_{f_2 \rightarrow c}(c)$$

$$\mu_{b \rightarrow f_1}(b) = \prod_{g \in \{ne(b) \setminus f_1\}} \mu_{g \rightarrow x}(x)$$

$$= \prod_{g \in \{f_2\}} \mu_{g \rightarrow x}(x)$$

$$= \mu_{f_2 \rightarrow b}(b) = 1.2b + 0.7$$

$$\mu_{b \rightarrow f_2}(b) = \prod_{g \in \{ne(b) \setminus f_2\}} \mu_{g \rightarrow x}(x)$$

$$= \prod_{g \in \{f_1\}} \mu_{g \rightarrow x}(x)$$

$$= \mu_{f_1 \rightarrow b}(b)$$

$$= 1$$

3. Factor-to-Variable

$$\begin{aligned}
 \mu_{f_2 \rightarrow b}(b) &= \sum_{c,d} f_2(b,c,d) \prod_{y \in \{ne(f_2) \setminus b\}} \mu_{y \rightarrow f_2}(y) \\
 &= \sum_{c,d} f_2(b,c,d) \prod_{y \in \{c,d\}} \mu_{y \rightarrow f_2}(y) \\
 &= \sum_{c,d} f_2(b,c,d) \times \mu_{c \rightarrow f_2}(c) \times \mu_{d \rightarrow f_2}(d) \\
 &= \sum_{c,d} 0.6b + 0.2c + 0.3d \times [c=1] \times 1
 \end{aligned}$$

c	d	
0	0	
0	1	
1	0	$0.6b + 0.2$
1	1	$0.6b + 0.5$

$= 1.2b + 0.7$

$$\begin{aligned}
 \mu_{f_1 \rightarrow b}(b) &= \sum_a f_1(a,b) \prod_{y \in \{ne(f_1) \setminus a\}} \mu_{y \rightarrow f_1}(y) \\
 &= \sum_a f_1(a,b) \prod_{y \in \{a\}} \mu_{y \rightarrow f_1}(y) \\
 &= \sum_a f_1(a,b) \mu_{a \rightarrow f_1}(b) \\
 &= 1
 \end{aligned}$$

$$\mu_{f_1 \rightarrow a}(a) = \sum_{b,c,d} f_1(a,b) \times \prod_{y \in \{ne(f_1) \setminus a\}} \mu_{y \rightarrow f_1}(y)$$

$$= \sum_{b,c,d} f_1(a,b) \times \prod_{y \in \{b\}} \mu_{y \rightarrow f_1}(y)$$

b
0
1

$$= \sum_b f_1(a, b) \times \mu_{b \rightarrow f_1}(b) \quad \text{Case } a=0: \\ 0.7$$

$$= 1.2a + 0.7 \leftarrow \text{as a func } \rightarrow \quad \text{Case } a=1: \\ 1.2 + 0.7 = 1.9$$

$$\mu_{f_2 \rightarrow c}(c) = \sum_{b, d} f_2(b, c, d) \prod_{y \in \{ne(f_2)/c\}} \mu_{y \rightarrow f_2}(y)$$

$$= \sum_{b, d} f_2(b, c, d) \prod_{y \in \{b, d\}} \mu_{y \rightarrow f_2}(y)$$

$$= \sum_{b, d} f_2(b, c, d) \times \mu_{b \rightarrow f_2}(b) \times \mu_{d \rightarrow f_2}(d)$$

$$= 0.8c + 1.8 \quad \times 1 \times 1$$

b	d	
0	0	0.2c
0	1	0.2c + 0.3
1	0	0.6 + 0.2c
1	1	0.2c + 0.9

$$\mu_{f_2 \rightarrow d}(d) = \sum_{b, c} f_2(b, c, d) \prod_{y \in \{ne(f_2) \setminus d\}} \mu_{y \rightarrow f_2}(y)$$

$$= \sum_{b, c} f_2(b, c, d) \prod_{y \in \{b, c\}} \mu_{y \rightarrow f_2}(y)$$

$$= \sum_{b, c} f_2(b, c, d) \times \mu_{b \rightarrow f_2}(b) \times \mu_{c \rightarrow f_2}(c)$$

$$= \sum_{b, c} 0.6b + 0.2c + 0.3d \times 1 \times [c=1]$$

$$= 0.6d + 1.0$$

b	c	
0	0	$0.3d + 0.2$
0	1	
1	0	$0.3d + 0.8$
1	1	

4. Calculate Marginals

$$p(x) \propto \prod_{f \in \text{ne}(x)} \mathcal{U}_{f \rightarrow x}(x)$$

1. $p(a) \propto \prod_{f \in \text{ne}(a)} \mathcal{U}_{f \rightarrow a}(a)$

$$\propto \prod_{f \in \{f_1\}} \mathcal{U}_{f \rightarrow a}(a)$$

$$\propto \mathcal{U}_{f_1 \rightarrow a}(a) \propto 1.2a + 0.7 \quad \begin{cases} a=0 & 0.7 / (0.7 + 1.9) = 0.269 \\ a=1 & 1.9 / (0.7 + 1.9) = 0.73 \end{cases}$$

2. $p(b) \propto \prod_{f \in \text{ne}(b)} \mathcal{U}_{f \rightarrow b}(b)$

$$\propto \prod_{f \in \{f_1, f_2\}} \mathcal{U}_{f \rightarrow b}(b)$$

$$\propto \mathcal{U}_{f_1 \rightarrow b}(b) \times \mathcal{U}_{f_2 \rightarrow b}(b)$$

$$\propto 1.2b + 0.7 \quad \begin{cases} b=0 & 0.7 / (0.7 + 1.9) = 0.269 \\ b=1 & 1.9 / (0.7 + 1.9) = 0.73 \end{cases}$$

$$3. p(c) \propto \prod_{f \in \text{ne}(c)} \mu_{f \rightarrow c}(c)$$

$$\propto \prod_{f \in \{f_2, f_3\}} \mu_{f \rightarrow c}(c)$$

$$\propto \mu_{f_2 \rightarrow c}(c) \times \mu_{f_3 \rightarrow c}(c)$$

$$\propto 0.8c + 1.0 \times [c=1]$$

$$\propto \begin{cases} c=0 & 0 \\ c=1 & 0.8 + 1.0 / (0.8 + 1.0) = 1 \end{cases}$$

$$\propto [c=1]$$

$$4. p(d) \propto \prod_{f \in \text{ne}(d)} \mu_{f \rightarrow d}(d)$$

$$\propto \prod_{f \in \{f_2\}} \mu_{f \rightarrow d}(d)$$

$$\propto \mu_{f_2 \rightarrow d}(d)$$

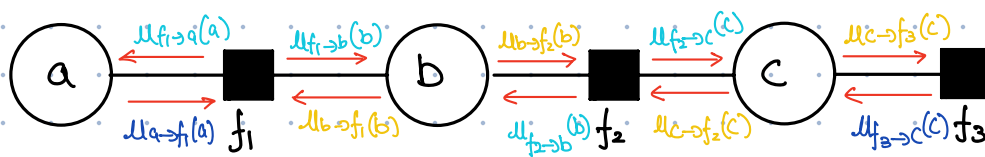
$$\propto 0.6d + 1.0 \begin{cases} d=0 & 1 / (1 + 1.6) = 0.38 \\ d=1 & 1.6 / (1 + 1.6) = 0.615 \end{cases}$$

(C) **Maximum-A-Posteriori.** Consider a distribution defined over binary variables $a, b, c \in \{0, 1\}$:

$$p(a, b, c) = \frac{1}{Z} f_1(a, b) f_2(b, c) f_3(c)$$

The factors are given as

$$f_1(a, b) = \begin{pmatrix} 0.4 & 0.15 \\ 0.12 & 0.25 \end{pmatrix}, \quad f_2(b, c) = \begin{pmatrix} 0.3 & 0.5 \\ 0.12 & 0.22 \end{pmatrix}, \quad f_3(c) = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}$$



1. Initialization:

$$\mu_{f_3 \rightarrow c}(c) = f_3(c) \quad \mu_{a \rightarrow f_1}(a) = 1$$

2. Variable-to-factor:

$$\mu_{c \rightarrow f_3}(c) = \prod_{g \in \{ne(c) \setminus f_3\}} \mu_{g \rightarrow c}(c)$$

$$= \prod_{g \in \{f_2\}} \mu_{g \rightarrow c}(c)$$

$$= \mu_{f_2 \rightarrow c}(c)$$

$$\mu_{c \rightarrow f_2}(c) = \prod_{g \in \{ne(c) / f_2\}} \mu_{g \rightarrow c}(c)$$

$$= \prod_{g \in \{f_3\}} \mu_{g \rightarrow c}(c)$$

$$= \mu_{f_3 \rightarrow c}(c)$$

$$= f_3(c)$$

$$\mu_{b \rightarrow f_2}(b) = \prod_{g \in \{ne(b) \setminus f_2\}} \mu_{g \rightarrow b}(b)$$

$$= \prod_{g \in \{f_1\}} \mu_{g \rightarrow b}(b)$$

$$= \mu_{f_1 \rightarrow b}(b)$$

$$\mu_{b \rightarrow f_1}(b) = \prod_{g \in \{ne(b) \setminus f_2\}} \mu_{g \rightarrow b}(b)$$

$$= \prod_{g \in \{f_2\}} \mu_{g \rightarrow b}(b)$$

$$= \mu_{f_2 \rightarrow b}(b)$$

3. Factor-to-Variable.

$$\mu_{f_2 \rightarrow c}(c) = \max_b f_2(b, c) \prod_{g \in \{ne(f_2) \setminus c\}} \mu_{g \rightarrow f_2}(g)$$

$$= \max_b f_2(b, c) \prod_{g \in \{b\}} \mu_{g \rightarrow f_2}(g)$$

$$= \max_b f_2(b, c) \mu_{b \rightarrow f_2}(b)$$

$$= \max_b f_2(b, c) \mu_{f_1 \rightarrow b}(b)$$

$$\begin{cases} c=0 & \max[0.3 \ 0.12] = 0.3 \\ c=1 & \max[0.5 \ 0.22] = 0.5 \end{cases} \quad \begin{cases} b=0 & \max[0.4 \ 0.12] = 0.4 \\ b=1 & \max[0.15 \ 0.25] = 0.25 \end{cases}$$

$$\begin{cases} c=0 & 0.12 \\ c=1 & 0.2 \end{cases}$$

$$\mu_{f_2 \rightarrow b}(b) = \max_c f_2(b, c) \prod_{y \in \{ne(f_2) \setminus b\}} \mu_{y \rightarrow f_2}(y)$$

$$= \max_c f_2(b, c) \prod_{y \in \{c\}} \mu_{y \rightarrow f_2}(y)$$

$$= \max_c f_2(b, c) \mu_{c \rightarrow f_2}(c)$$

$$= \max_c f_2(b, c) \quad f_3(c)$$

$$= \begin{cases} b=0 & \max[0.3 \times 0.6 \quad 0.5 \times 0.4] = 0.2 \\ b=1 & \max[0.12 \times 0.6 \quad 0.22 \times 0.4] = 0.088 \end{cases}$$

$$\mu_{f_1 \rightarrow b}(b) = \max_a f_1(a, b) \prod_{y \in \{ne(f_1) \setminus b\}} \mu_{y \rightarrow f_1}(y)$$

$$= \max_a f_1(a, b) \prod_{y \in \{a\}} \mu_{y \rightarrow f_1}(y)$$

$$= \max_a f_1(a, b) \mu_{a \rightarrow f_1}(b)$$

$$= \max_a f_1(a, b) \times 1$$

$$\begin{cases} b=0 & \max[0.4 \quad 0.12] = 0.4 \\ b=1 & \max[0.15 \quad 0.25] = 0.25 \end{cases}$$

$$\mu_{f_1 \rightarrow a}(a) = \max_b f_1(a, b) \prod_{y \in \{ne(f_1)/a\}} \mu_{g \rightarrow f_1}(y)$$

$$= \max_b f_1(a, b) \prod_{y \in \{b\}} \mu_{g \rightarrow f_1}(y)$$

$$= \max_b f_1(a, b) \mu_{b \rightarrow f_1}(b)$$

$$= \max_b f_1(a, b) \mu_{f_2 \rightarrow b}(b)$$

$$= \begin{cases} a=0 & \max[0.4 \ 0.15] = 0.4 \\ a=1 & \max[0.12 \ 0.25] = 0.25 \end{cases} \quad \begin{cases} b=0 & 0.2 \\ b=1 & 0.088 \end{cases}$$

$$= \begin{cases} a=0 & 0.08 \\ a=1 & 0.022 \end{cases}$$

4. Max

$$x^* = \underset{x}{\operatorname{argmax}} \sum_{g \in \{ne(x)\}} \mu_{g \rightarrow x}(x)$$

$$a^* = \underset{a}{\operatorname{argmax}} \sum_{g \in \{a\}} \mu_{g \rightarrow a}(a)$$

$$= \underset{a}{\operatorname{argmax}} \mu_{f_1 \rightarrow a}(a)$$

$$= 0$$

$$b^* = \argmax_b \sum_{g \in \{ne(b)\}} \mu_{g \rightarrow b}(b)$$

$$= \argmax_b \sum_{g \in \{f_1, f_2\}} \mu_{g \rightarrow b}(b)$$

$$= \argmax_b \mu_{f_1 \rightarrow b} \mu_{f_2 \rightarrow b}$$

$$\argmax_b \begin{cases} b=0 & 0.4 \\ b=1 & 0.25 \end{cases} \begin{cases} b=0 & 0.2 \\ b=1 & 0.088 \end{cases}$$

$$\argmax_b \begin{cases} b=0 & 0.08 \\ b=1 & 0.022 \end{cases}$$

$$= 0$$

$$c^* = \argmax_c \sum_{g \in \{ne(c)\}} \mu_{g \rightarrow c}(c)$$

$$= \argmax_c \sum_{g \in \{f_2, f_3\}} \mu_{g \rightarrow c}(c)$$

$$= \argmax_c \mu_{f_2 \rightarrow c}(c) \mu_{f_3 \rightarrow c}(c)$$

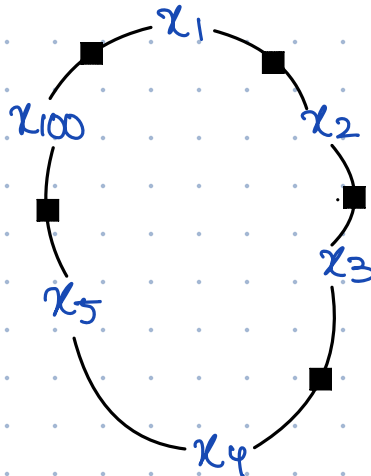
$$= \argmax_c \begin{cases} c=0 & 0.12 \\ c=1 & 0.2 \end{cases} \begin{cases} c=0 & 0.6 \\ c=1 & 0.4 \end{cases}$$

$$= \operatorname{argmax}_c \begin{cases} c=0 & 0.072 \\ c=1 & 0.08 \end{cases}$$

$$= 1$$

(d) **Loop Cut.** Consider a pairwise Markov Random Field defined on a ring (a chain that is connected at both ends) with 100 binary variables:

$$p(x) = \phi_0(x_1, x_{100}) \prod_{i=1}^{99} \phi_i(x_i, x_{i+1})$$



If we condition on x_1 , the circle would become a straight line

$$\operatorname{argmax}_{x_2, x_3, \dots, x_{100}} p(x | x_1 = 1)$$

$$\operatorname{argmax}_{x_2, x_3, \dots, x_{100}} p(x | x_1 = 0)$$

Now, simply choose the one with higher value when plugging back to $p(x)$.

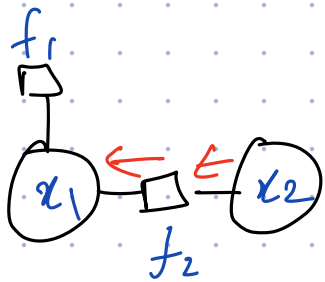
Coding:

Q1.

$$\mu_{f \rightarrow x}(x) = \max_{x \setminus x} f(x_f) \prod_{y \in \{ne(f) \setminus x\}} \mu_{y \rightarrow f}(y)$$

↳ No neighbours.

$$\mu_{f \rightarrow x}(x) = \max_{x \setminus x} f(x_f)$$

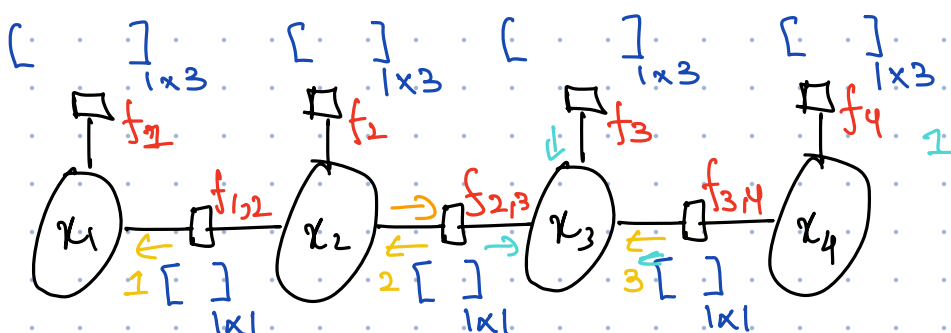


$$\sum_y f_2(x, y) \prod_{y \in \{ne(f_2) \setminus x\}} \mu_{y \rightarrow f_2}(y)$$

$$\mu_{x_2 \rightarrow f_2}(x_2)$$

$$f(x_i) = \begin{bmatrix} \text{Var 1} \\ \text{Var 2} \\ \vdots \end{bmatrix} \quad f(x_i, x_{i+1}) = \begin{bmatrix} \# \\ \# \\ \# \end{bmatrix}$$

msg
9x3



$$\begin{aligned} \mu_{f_{3,4} \rightarrow x_3} &= \max_4 f_{3,4} \prod_{y \in \{ne(f_{3,4})/x_3\}} \mu_{y \rightarrow f_{3,4}}(y) \\ &= \max_4 f_{3,4} \mu_{x_4 \rightarrow f_{3,4}}(x_4) \end{aligned}$$

$$\begin{aligned} \mu_{x_4 \rightarrow f_{3,4}}(x_4) &= \mu_{f_4 \rightarrow x_4} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \mu_{f_{2,3} \rightarrow x_2} &= \max_3 f_{2,3} \prod_{y \in \{ne(f_{2,3})/x_2\}} \mu_{y \rightarrow f_{2,3}}(y) \\ &= \max_3 f_{2,3} \mu_{x_3 \rightarrow f_{2,3}} \end{aligned}$$

$$\begin{aligned} \mu_{x_3 \rightarrow f_{2,3}} &= \prod_{y \in \{ne(x_3) \setminus f_{2,3}\}} \mu_{y \rightarrow x_3}(x_3) \\ &= \mu_{f_3 \rightarrow x_3}(x_3) \times \mu_{f_{3,4} \rightarrow x_3}(x_3) \end{aligned}$$

$$\max_3 f_{2,3} \times \mu_{f_3 \rightarrow x_3}(x_3) \times \mu_{f_{3,4} \rightarrow x_3}(x_3)$$

pairwise unaxxy incoming msg
 axis = 1

Max-marginals:

$$x_1^* = \mu_{f_1 \rightarrow x_1}(x_1) * \mu_{f_{1,2} \rightarrow x_1}(x_1)$$

$$\underline{\mu_{f_{1,2} \rightarrow x_1}(x_1)} = \max_2 f_{1,2} \mu_{x_2 \rightarrow f_{1,2}}(x_2)$$

$$\underline{\mu_{x_2 \rightarrow f_{1,2}}(x_2)} = \mu_{f_2 \rightarrow x_2}(x_2) * \mu_{f_{2,3} \rightarrow x_2}(x_2)$$

$$x_1^* = \mu_{f_1 \rightarrow x_1}(x_1) * \max_2 f_{1,2} * \mu_{f_2 \rightarrow x_2}(x_2) * \mu_{f_{2,3} \rightarrow x_2}(x_2)$$

$$\text{unary}[i] * \max_2 [\text{pairwise} * \text{unary}[i+1]] * \text{msg}[i]$$

$$x_2^* = \mu_{f_{1,2} \rightarrow x_2}(x_2) * \mu_{f_2 \rightarrow x_2}(x_2) * \mu_{f_{2,3} \rightarrow x_2}(x_2)$$

$$\mu_{f_{1,2} \rightarrow x_2}(x_2) = \max_1 f_{1,2} \mu_{x_1 \rightarrow f_{1,2}}(x_1)$$

$$\mu_{x_1 \rightarrow f_{1,2}}(x_1) = \mu_{x_2 \rightarrow f_{1,2}}(x_2)$$

dp[i-1]

$$x_2^* = \mu_{f_2 \rightarrow x_2}(x_2) * \mu_{f_{2,3} \rightarrow x_2}(x_2) \max_1 f_{1,2} \mu_{x_2 \rightarrow f_{1,2}}(x_2)$$

$$\text{unary}[i] * \text{msg}[i] * \max [\text{pairwise} * \text{dp}[i-1]]$$

$$\mu_3^* = \mu_{f_3 \rightarrow x_3}(x_3) * \mu_{f_{3,4} \rightarrow x_3}(x_3) * \mu_{f_{2,3} \rightarrow x_3}(x_3)$$

✓ ✓

$$\mu_{f_{2,3} \rightarrow x_3}(x_3) = \max_2 f_{2,3} \mu_{x_2 \rightarrow f_{2,3}}(x_2)$$

$$\mu_{x_2 \rightarrow f_{2,3}}(x_2) = \mu_{f_{1,2} \rightarrow x_2}(x_2) * \mu_{f_2 \rightarrow x_2}(x_2)$$

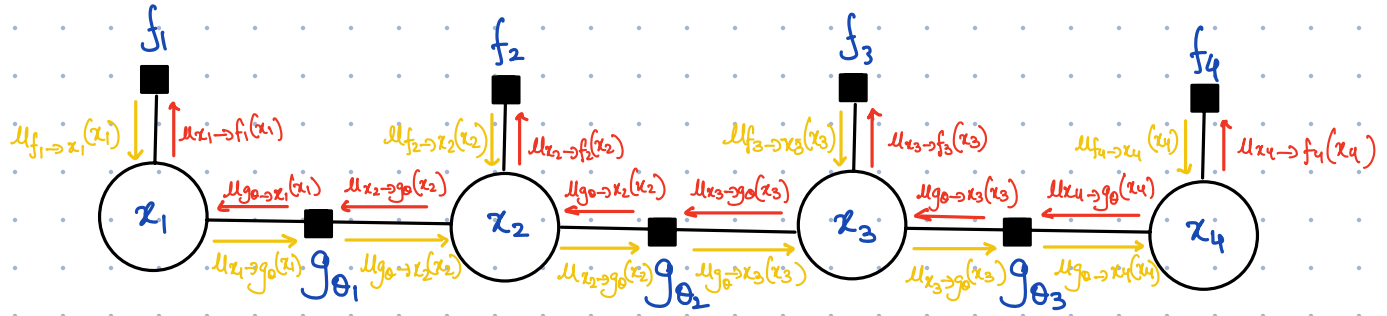
$$\mu_3^* = \mu_{f_3 \rightarrow x_3}(x_3) * \mu_{f_{3,4} \rightarrow x_3}(x_3) \max_2 f_{2,3} \mu_{f_{1,2} \rightarrow x_2}(x_2) * \mu_{f_2 \rightarrow x_2}(x_2)$$

$$\text{unary}[i] * \text{msg}[i] \max_2 \text{pairwise} * \text{dp}[i] * \text{unary}[i-1]$$

$$\mu_4^* =$$

I give up!

Lets try again:



Left-Right Messages:

$$u_{f_1 \rightarrow x_1}(x_1) = 1$$

$$u_{x_1 \rightarrow g_\theta}(x_1) = \prod_{g \in \{ne(x_1)\}} u_{g \rightarrow x_1}(x_1)$$

$$= \prod_{g \in \{f_1\}} u_{g \rightarrow x_1}(x_1)$$

$$= u_{f_1 \rightarrow x_1}(x_1)$$

$$u_{g_\theta \rightarrow x_2} = \max_x \left[g_\theta(x_1, x_2) \prod_{g \in \{ne(g_\theta)\}} u_{y \rightarrow f(y)} \right]$$

$$\max \left[g_\theta(x_1, x_2) \prod_{g \in \{x_1\}} u_{y \rightarrow f(y)} \right]$$

$$\max \left[g_\theta(x_1, x_2) u_{x_1 \rightarrow g_\theta}(x_2) \right]$$

$$\mu_{x_2 \rightarrow g_{\theta_2}}(x_1) = \prod_{g \in \{ne(x_2)\}} \mu_{g \rightarrow x_2}(x_2)$$

$$= \prod_{g \in \{f_2, g_{\theta_1}\}} \mu_{g \rightarrow x_2}(x_2)$$

$$= \mu_{f_2 \rightarrow x_2}(x_2) \times \mu_{g_{\theta_1} \rightarrow x_2}(x_2)$$

$$\mu_{f_2 \rightarrow x_2}(x_2) = \max_{x_1} \left[f_2(x_2) \prod_{y \in \{ne(f_2)\}} \mu_{y \rightarrow f_2}(y) \right]$$

$$\max \left[f_2(x_2) \right]$$

$$\mu_{g_{\theta_2} \rightarrow x_3}(x_3) = \max_{x_2} \left[g_{\theta} \prod_{y \in \{ne(g_{\theta})\}} \mu_{y \rightarrow g_{\theta}}(y) \right]$$

$$\max \left[g_{\theta} \prod_{y \in \{x_2\}} \mu_{y \rightarrow g_{\theta}}(y) \right]$$

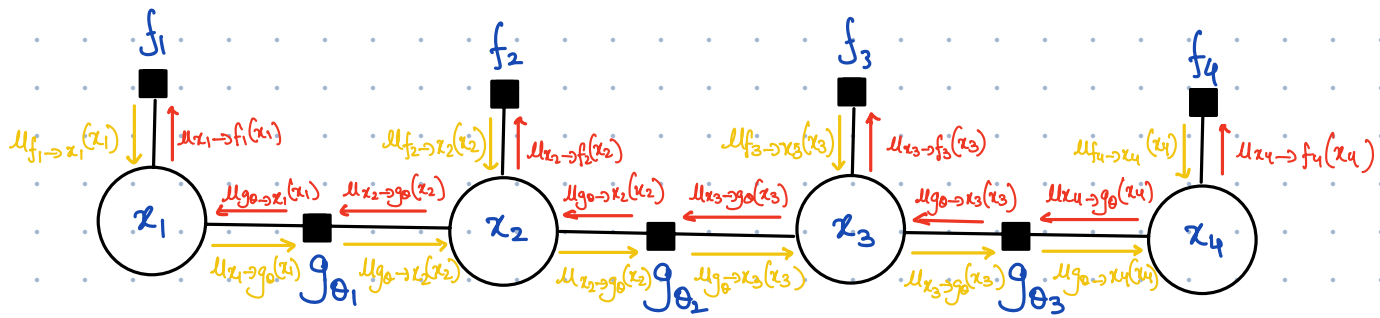
$$\max \left[g_{\theta} * \mu_{x_2 \rightarrow g_{\theta}}(x_2) \right]$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}_{3 \times 1}$$

[

$$\mu_{x_2 \rightarrow f_2}(x_2) = \mu_{g_{\theta} \rightarrow x_2} * \mu_{g_{\theta} \rightarrow x_2}$$



Right-left Messages:

$$\begin{aligned} \mu_{x_4 \rightarrow f_4}(x_4) &= \prod_{g \in \{ne(x_4)\}} \mu_{g \rightarrow x_4}(x_4) \\ &= \mu_{g_{\theta_3} \rightarrow x_4}(x_4) \end{aligned}$$

$$\begin{aligned} \mu_{x_4 \rightarrow g_{\theta_3}}(x_4) &= \prod_{g \in \{ne(x_4)\}} \mu_{g \rightarrow x_4}(x_4) \\ &= \prod_{g \in \{f_4\}} \mu_{g \rightarrow x_4}(x_4) \\ &= \mu_{f_4 \rightarrow x_4}(x_4) = f_4(x_4) \end{aligned}$$

$$\begin{aligned} \mu_{g_{\theta_3} \rightarrow x_3}(x_3) &= \max_{x_4} \left[g_{\theta_3} \prod_{y \in \{ne(g_{\theta_3})/x_3\}} \mu_{y \rightarrow g_{\theta_3}}(y) \right] \\ &= \max_{x_4} \left[g_{\theta_3} \prod_{y \in \{x_4\}} \mu_{y \rightarrow g_{\theta_3}}(y) \right] \\ &= \max_{x_4} [g_{\theta_3} * \mu_{x_4 \rightarrow g_{\theta_3}}] \end{aligned}$$

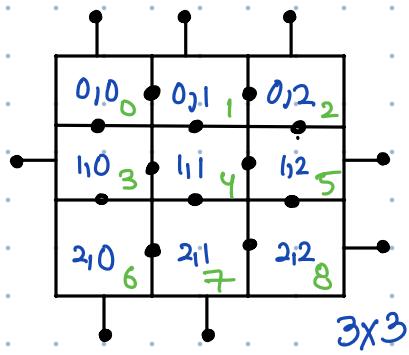
$$\begin{aligned}
 \mu_{x_3 \rightarrow f_3}(x_3) &= \prod_{g \in \{ne(x_3)\}} \mu_{g \rightarrow x_3}(x_3) \\
 &= \prod_{g \in \{g_{\theta_2}, g_{\theta_3}\}} \mu_{g \rightarrow x_3}(x_3) \\
 &= \mu_{g_{\theta_2} \rightarrow x_3} * \mu_{g_{\theta_3} \rightarrow x_3}
 \end{aligned}$$

$$\begin{aligned}
 \mu_{x_3 \rightarrow g_{\theta_2}}(x_3) &= \prod_{g \in \{ne(x_3)\}} \mu_{g \rightarrow x_3}(x_3) \\
 &= \prod_{g \in \{g_{\theta_3}, f_3\}} \mu_{g \rightarrow x_3}(x_3) \\
 &= \mu_{g_{\theta_3} \rightarrow x_3}(x_3) * \mu_{f_3 \rightarrow x_3}(x_3)
 \end{aligned}$$

$$\begin{aligned}
 \mu_{g_{\theta_2} \rightarrow x_2}(x_2) &= \max_{x_3} \left[g_{\theta_2} \prod_{y \in \{ne(g_{\theta_2})\}} \mu_{y \rightarrow g_{\theta_2}}(y) \right] \\
 &= \max_{x_3} \left[g_{\theta_2} \prod_{y \in \{x_3\}} \mu_{y \rightarrow g_{\theta_2}}(y) \right] \\
 &= \max_{x_3} \left[g_{\theta_2} * \mu_{x_3 \rightarrow g_{\theta_2}}(x_3) \right]
 \end{aligned}$$

$$\mu_{x_1 \rightarrow f_1}(x_1) = \mu_{g_{\theta} \rightarrow x_1}(x_1)$$

Task 02:



for example,

pixel $(1,1)$

pairwise:

